ENETERPRISE CONNECTION BUILDING USING CISCO

REQUIREMENTS:

- ISP ROUTER
- EDGE ROUTER
- CORE ROUTER
- SWITCHES
- SERVER
- PCS

IMPLEMENTING THE CONCEPT STEP BY STEP

STEP 1:

CONFIGURING ISP ROUTER:

```
Router>EN
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giaga0/0/0
                         ^
% Invalid input detected at '^' marker.

Router(config)#interface giga0/0/0
Router(config-if)#ip add 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

CONFIGURING EDGE ROUTER:

ISP TO EDGE

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/0
Router(config-if)#ip add 200.1.1.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

EDGE TO CORE

```
Router>
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

EDGE TO DMZ

```
Router(config)#interface giga0/0/2
Router(config-if)#ip add 172.16.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/2, changed state to up

Router(config-if)#exit
Router(config)#
```

CORE TO EDGE

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/0
Router(config-if)#ip add 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

CREATING VLAN CONFIGURATION LOGICAL SEPARATION

STEP 1 :

CONFIGURING ROUTER

```
Router(config-if)#exit
Router(config)#interface giga0/0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#exit
Router(config)#interface giga0/0/1.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip add 192.168.1.10 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface giga0/0/1.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1.20, changed state to
up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip add 192.168.2.10 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

STEP 2:

CONFIGURING SWITCH

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#
```

OUTPUT :

```
C:\>ping 192.168.2.30

Pinging 192.168.2.30 with 32 bytes of data:

Reply from 192.168.2.30: bytes=32 time<1ms TTL=127
Reply from 192.168.2.30: bytes=32 time<1ms TTL=127
Reply from 192.168.2.30: bytes=32 time<1ms TTL=127
Reply from 192.168.2.30: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

IMPLEMENTING OSPF

(OPEN SHORTEST PATH FINDER)

“DYNAMIC ROUTING”

IMPLEMENTING OSPF

CORE :

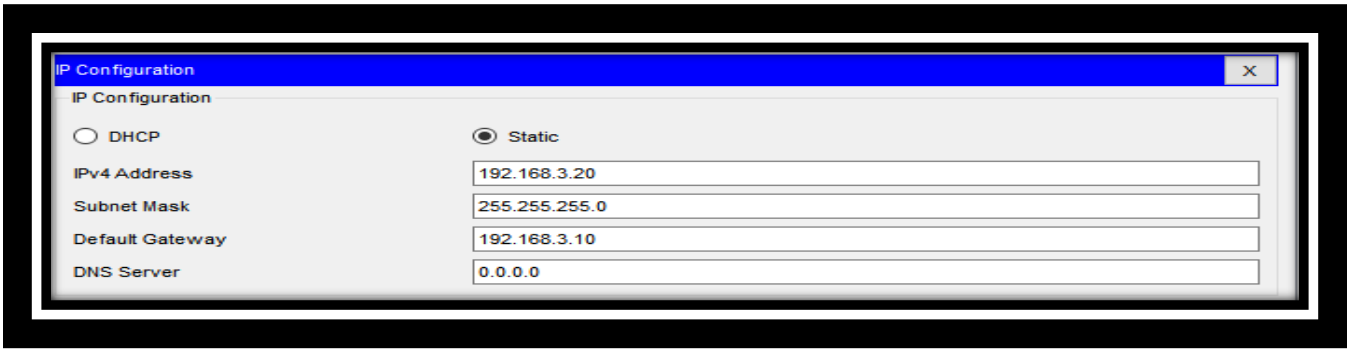
```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#exit
```

EDGE:

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
00:04:27: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.10 on GigabitEthernet0/0/1 from
LOADING to FULL, Loading Done

Router(config-router)#exit
```

CONFIGURING DMZ :



OUTPUT FOR DMZ WORKS :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.20

Pinging 192.168.3.20 with 32 bytes of data:

Reply from 192.168.3.20: bytes=32 time<1ms TTL=126
Reply from 192.168.3.20: bytes=32 time<1ms TTL=126
Reply from 192.168.3.20: bytes=32 time<1ms TTL=126
Reply from 192.168.3.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```